

JUSTIN M. PARRA

Dunellen, NJ | (732) 772-5709 | jstnp18@gmail.com | linkedin.com/in/justin-parra/ | github.com/justintime1888

EDUCATION

Rutgers University – New Brunswick, NJ

Expected May 2027

Bachelor of Science in Electrical and Computer Engineering (ECE)

Coursework: Semiconductor Devices, Microelectronic Circuits, Signals & Systems, Control Systems, Materials Physics

ENGINEERING EXPERIENCE

LPT-KEYPAK

Tewksbury, NJ

Engineering Intern

June 2024 – August 2024

- Commissioned and validated automated packaging machinery for the Bill & Melinda Gates Foundation, boosting line throughput by 30% during pilot production qualification runs.
- Diagnosed electromechanical and process failure points across automated equipment using root-cause analysis (5-Whys, Pareto charts), reducing production defect rates by 15%.
- Formulated process validation protocols and authored Best-Known Methods (BKMs) and maintenance SOPs to standardize equipment integration and operator workflows.
- Validated sensor calibration curves, voltage tolerances, and relay timing margins using digital storage oscilloscopes and precision multimeters, ensuring strict quality compliance.

Rutgers Micromouse

New Brunswick, NJ

Lead Hardware & Firmware Engineer | Co-President

September 2024 – Present

- Designed and routed custom 4-layer PCBs in Altium Designer with ENIG surface finish, optimizing trace impedances, power planes, and coplanar ground shielding for signal integrity.
- Developed precision analog IR optical distance-sensing stages with active ambient-light filtering, improving surface detection repeatability to ± 2 mm at velocities up to 1.5 m/s.
- Characterized signal-to-noise ratios (SNR), rise times, and bus arbitration across 400 kHz I²C and SPI lines using mixed-signal oscilloscopes and logic analyzers, eliminating data dropouts.
- Directed 40+ student engineers across hardware and software sub-teams, leading design reviews, milestone tracking, and technical presentations to secure \$2,500+ in corporate sponsorship.

VEX Robotics – Rutgers IEEE

New Brunswick, NJ

Lead Electrical Engineer

September 2024 – Present

- Architected communication and distributed power delivery architectures, engineering RS-485 $\rightarrow$ UART $\rightarrow$ I²C transceiver networks between main controller and peripheral subsystems.
- Executed hardware-in-the-loop (HIL) stress testing and signal integrity validation under severe mechanical vibration using digital oscilloscopes to verify packet-loss rates $< 0.1\%$.

RELEVANT PROJECTS

Experimental Telemetry & Statistical Data Analysis Pipeline

Python, NumPy, SciPy, Matplotlib, Data Analysis & Visualization

- Built an automated statistical analysis pipeline in Python processing 100+ experimental test runs to evaluate transient settling metrics, overshoot variance, and parameter confidence intervals.
- Generated visual dashboards and distribution plots for experimental data interpretation, identifying process outliers and driving closed-loop system tuning.

Analog Filter & Transient Signal Conditioning Analyzer

LTspice, MATLAB, Signal Integrity, Circuit Modeling

- Simulated active 2nd-order RLC filter topologies and op-amp stages in LTspice, evaluating AC frequency responses, Bode phase margins, and transient damping factors.
- Designed hardware low-pass signal conditioning stages to suppress inductive switching transients and prevent ADC input distortion, verifying performance with benchtop instrumentation.

TECHNICAL SKILLS

Semiconductor & Packaging: Hybrid Bonding Concepts (Die-to-Wafer / W2W), Surface Preparation & Activation, Process Validation & BKMs, Defectivity Reduction, Process Uniformity, Cleanroom Protocols

Metrology & Characterization: Surface Profilometry & Roughness, AFM Principles, Optical Inspection, Contact Angle / Surface Energy Concepts, Bond Strength Testing, Oscilloscopes, Logic Analyzers

Data Analysis & Modeling: Python (NumPy, SciPy, Matplotlib), MATLAB, Design of Experiments (DOE), Statistical Process Control (SPC), Root-Cause Analysis (5-Whys), LTspice, Multisim

Hardware & Electronic Design: Altium Designer (4-Layer PCB Routing, ENIG Surface Finish), Analog Signal Conditioning, Sensor Calibration, Microcontrollers (ESP32-S3, STM32), C, C++ (C++11/17)